Personal Loan Campaign Project Machine Learning and Business Applications

By Rithvik Chandrupatla

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Problem Statement

- Given a varying customer base and a poor amount of asset customers, how can AllLife Bank leverage the financial data to expand upon the average customer's need for personal loan financing?

- This project aims to uncover statistical patterns amongst the financial attributes of customers to understand which types of customers are more likely to fall under the category of asset customers who may help improve loan business
- How can machine learning models help optimize target data for AllLife Bank?

Data Walkthrough

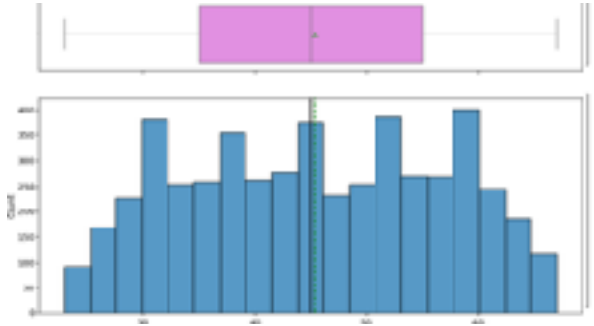
- [{"index":"ID","count":"5000.0","mean":2500.5,"std":1443.5200033252052,"min":"1.0","25%":"1250.75","50%":"2500.5","75%":"3750.25","max":"5000.0"},
{"index":"Age","count":"5000.0","mean":45.3384,"std":11.463165630542667,"min":"23.0","25%":"35.0","50%":"45.0","75%":"55.0","max":"67.0"},
{"index":"Experience","count":"5000.0","mean":20.1046,"std":11.467953681120553,"min":"-3.0","25%":"10.0","50%":"20.0","75%":"30.0","max":"43.0"},
{"index":"Income","count":"5000.0","mean":73.7742,"std":46.03372932108631,"min":"8.0","25%":"39.0","50%":"64.0","75%":"98.0","max":"224.0"},
{"index":"ZIPCode","count":"5000.0","mean":93169.257,"std":1759.4550864944745,"min":"90005.0","25%":"91911.0","50%":"93437.0","75%":"94608.0","max":"96651.0"},
{"index":"Family","count":"5000.0","mean":2.3964,"std":1.1476630455378525,"min":"1.0","25%":"1.0","50%":"2.0","75%":"3.0","max":"4.0"},
{"index":"CCAvg","count":"5000.0","mean":1.9379380000000002,"std":1.7476589800467675,"min":"0.0","25%":"0.7","50%":"1.5","75%":"2.5","max":"10.0"},
{"index":"Education","count":"5000.0","mean":1.881,"std":0.8398690826641989,"min":"1.0","25%":"1.0","50%":"2.0","75%":"3.0","max":"3.0"},
{"index":"Mortgage","count":"5000.0","mean":56.4988,"std":101.71380210211204,"min":"0.0","25%":"0.0","50%":"0.0","75%":"101.0","max":"635.0"},
{"index":"Personal_Loan","count":"5000.0","mean":0.096,"std":0.29462070577618,"min":"0.0","25%":"0.0","50%":"0.0","75%":"0.0","max":"1.0"},
{"index":"Securities_Account","count":"5000.0","mean":0.1044,"std":0.3058093260003261,"min":"0.0","25%":"0.0","50%":"0.0","75%":"0.0","max":"1.0"},
{"index":"CD_Account","count":"5000.0","mean":0.0604,"std":0.23825027311322797,"min":"0.0","25%":"0.0","50%":"0.0","75%":"0.0","max":"1.0"},
{"index":"Online","count":"5000.0","mean":0.5968,"std":0.490589334962671,"min":"0.0","25%":"0.0","50%":"1.0","75%":"1.0","max":"1.0"},
{"index":"CreditCard","count":"5000.0","mean":0.294,"std":0.45563748869492815,"min":"0.0","25%":"0.0","50%":"0.0","75%":"1.0","max":"1.0"}]

EDA & Pre-Processing

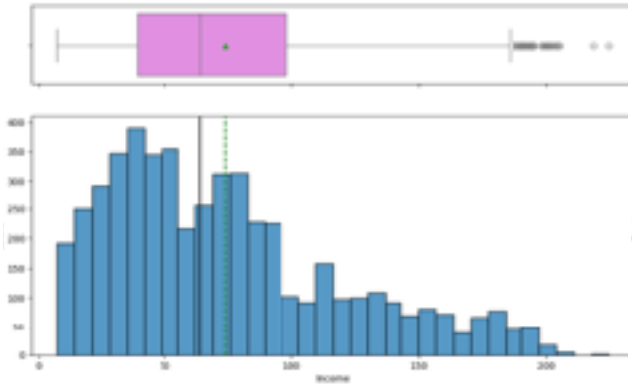
- Dropped the ID column since it would not contribute to data variability.
- Checked for Anomalous Values using corrections for all negative unique values (data[“Experience”].unique()) —> array([-1, -2, -3])

```
data["Experience"].replace(-1, 1, inplace=True)
data["Experience"].replace(-2, 2, inplace=True)
data["Experience"].replace(-3, 3, inplace=True)
```

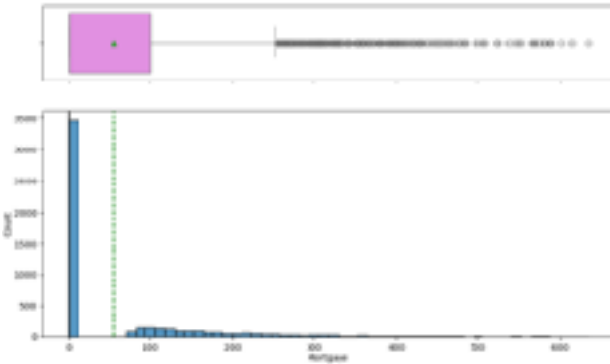
- Age Observations



Age Distributions are relatively Normal

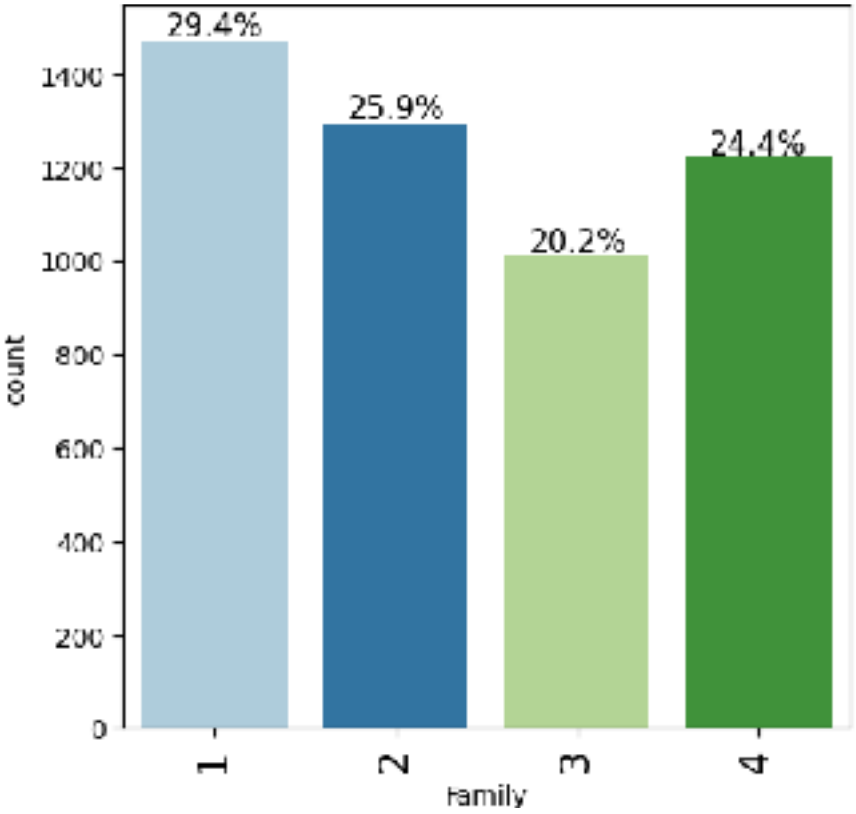


Income Distributions show a right skew with larger median

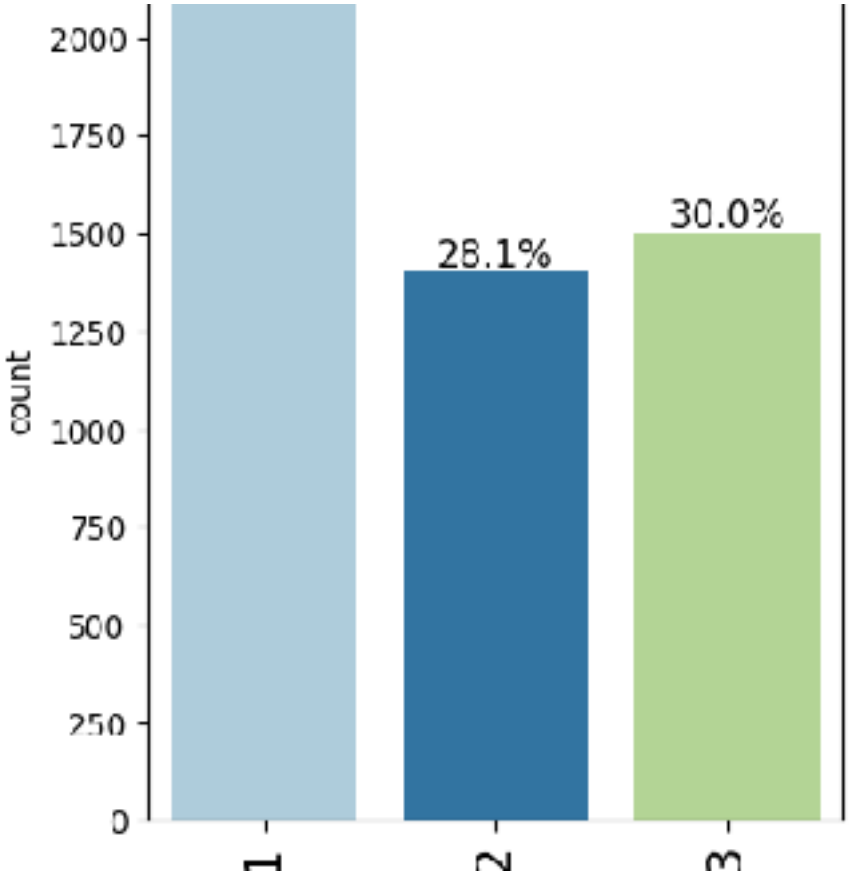


Mortgage distributions show a heavy skew that indicate high variability in loan fulfillment. Possible sector for asset customers

EDA continued

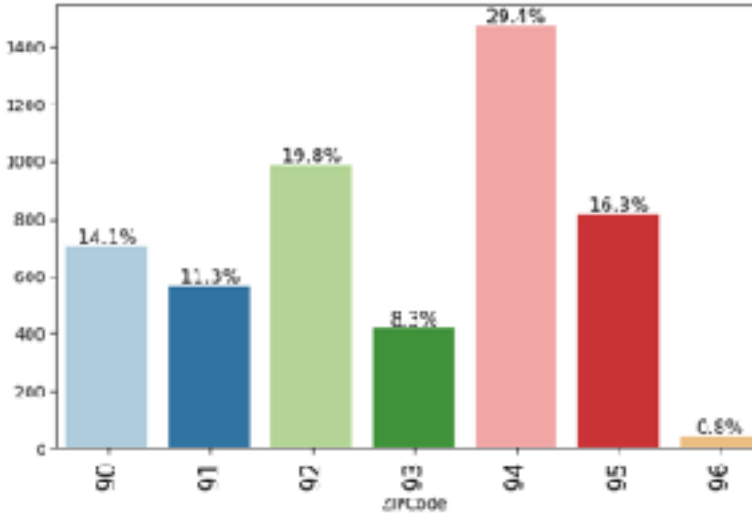


Family bar plots indicate larger proportion of single parent family homes that would have trouble repaying loans

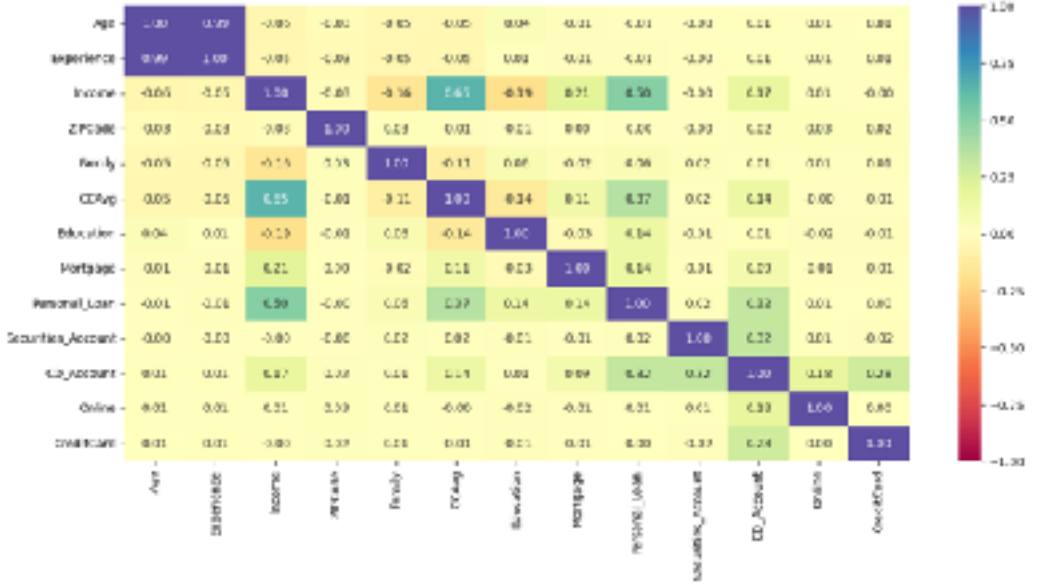


Larger proportions of households have only one educated family member, which distinguishes loan repayment options due to occupational struggles

EDA co. Bivariate Analysis



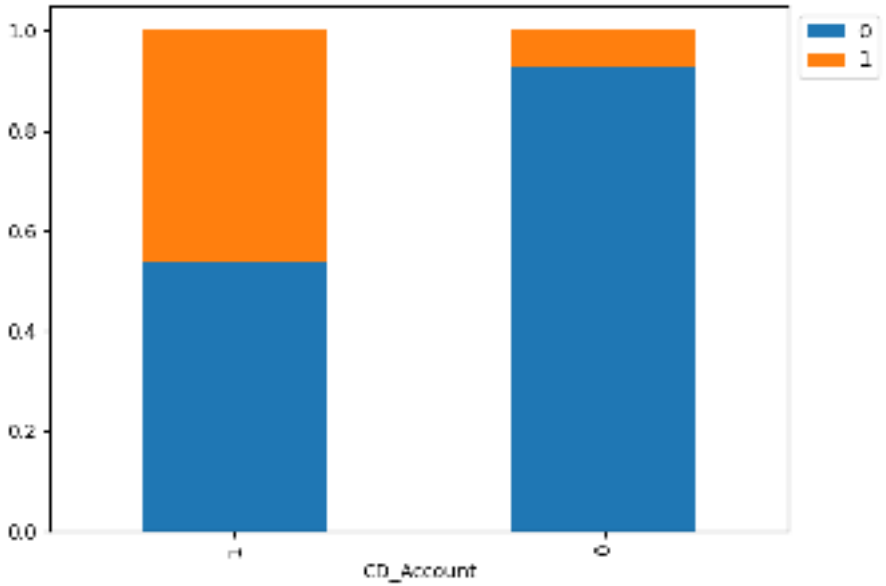
Zip Codes provide geographic distribution about the various underdeveloped areas where loan assets may be more lucrative.



Correlation Check indicates strong relations between CCAvg & Income, Personal_Loan & Income, and Education & Income. Personal Loan and CCAvg are variables with more impact on other independent factors such as Family and ZipCode.

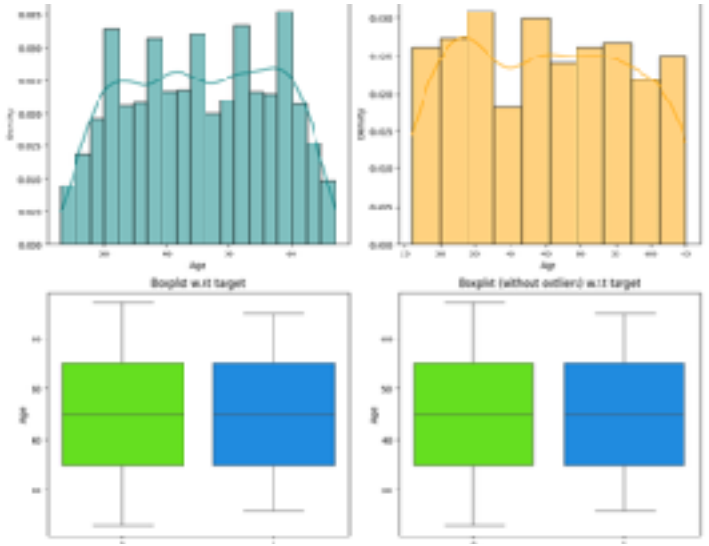
Bivariate Analysis cont.

- Most stacked bar plots correlating personal loan to financial characteristics yield no major patterns except for certificate of debit account holders



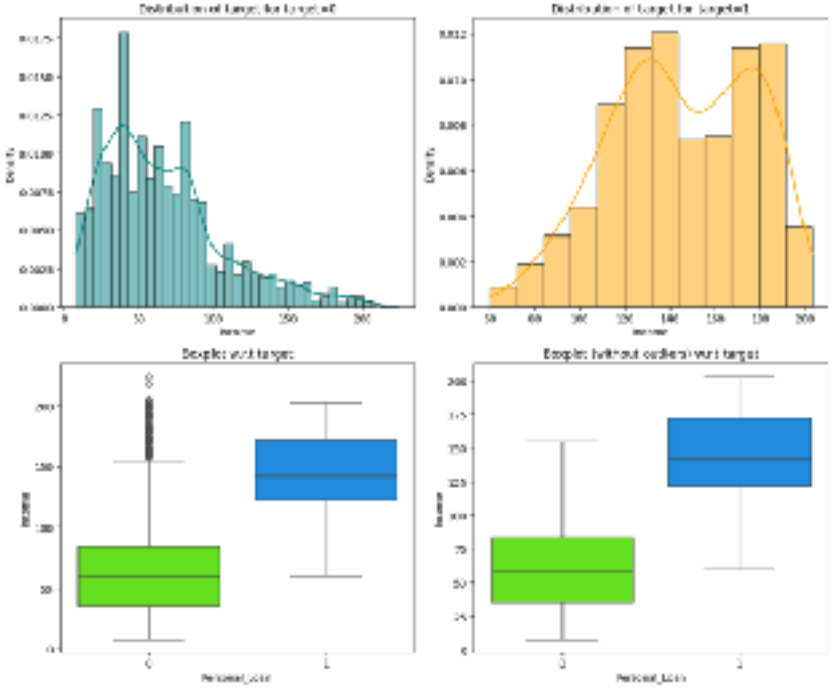
Stacked marplot of CD_account holders indicates more deposit savvy customers are less like to become credit loaners. 0 indicates no personal loan taken previously

Customer Loan Interest Varies Heavily with Age

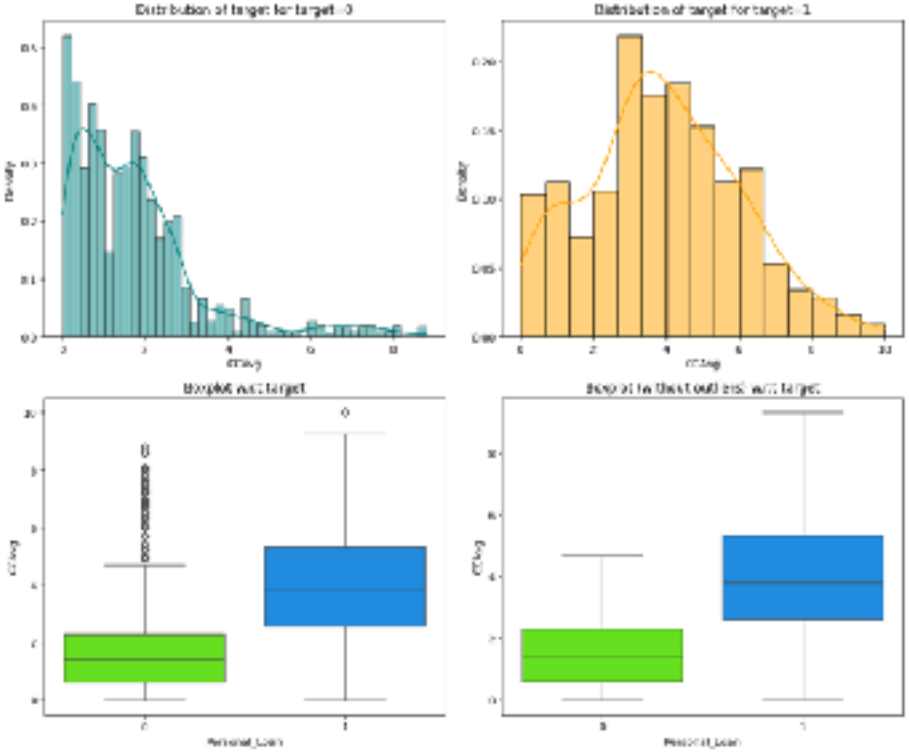


The target graphs for loaners and non loan customers showcases more prevalence in elder demographics

Bivariate Analysis



The right and left skew presented in these graphs indicate a high positive correlation with personal loan finance and income. This graph showcases how asset customers are more likely to have lower income status. Varying responses in distribution mean less precision



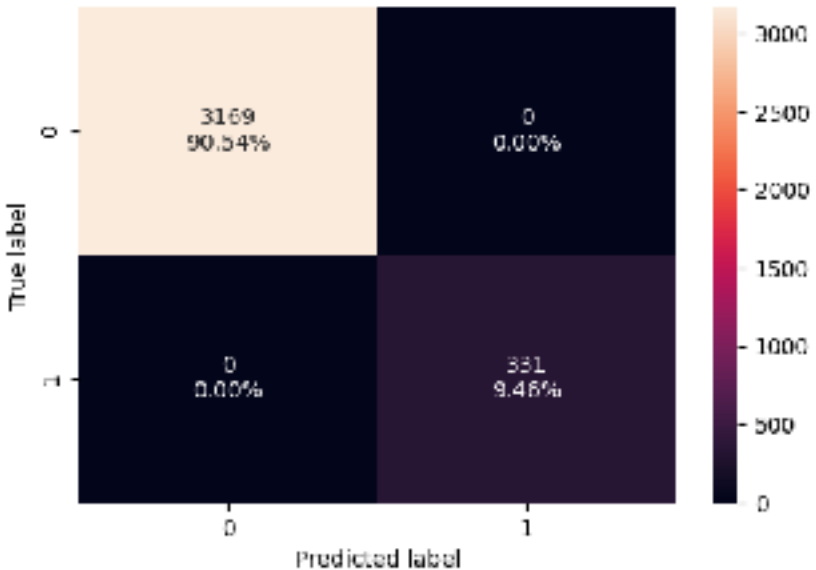
Personal Loan vs CCAvg showcases the same skewed distribution for both responses indicating stronger recall. More likely for creditors to pursue personal loan customers.

Data Preparation for Modeling

- Using Personal Loan Vs Experience data since it adheres to the uniform normal distribution rules
- No need for missing value treatment, verified original values using is.sum() and value_counts().
- Outlier Detection using df.quantile(%)

```
Shape of Training set :  (3500, 17)
Shape of test set :  (1500, 17)
Percentage of classes in training set:
Personal_Loan
0    0.905429
1    0.094571
Name: proportion, dtype: float64
Percentage of classes in test set:
Personal_Loan
0    0.900667
1    0.099333
Name: proportion, dtype: float64
```

Using model_performance_classification_sklearn

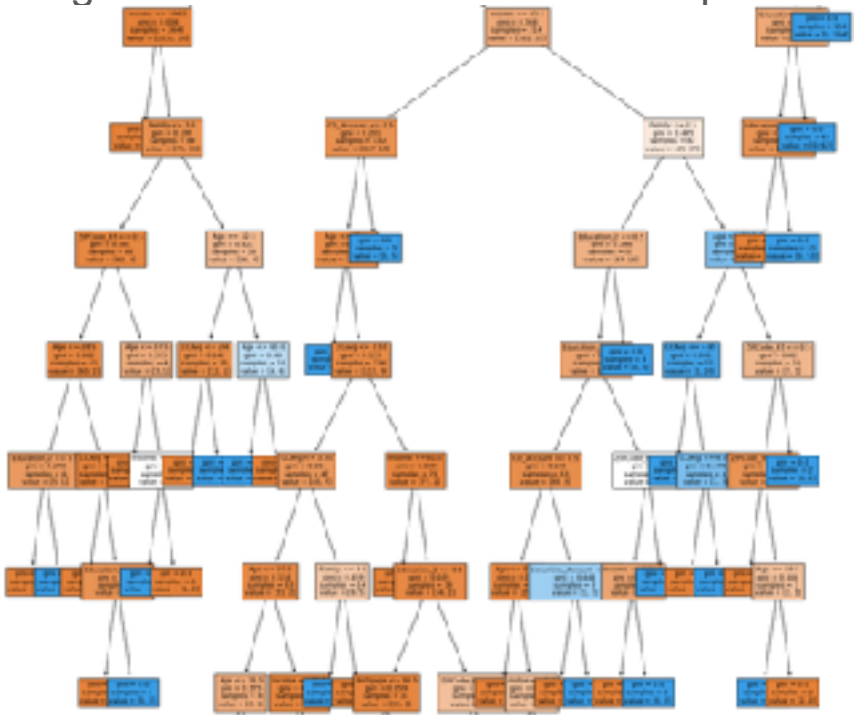


Confusion matrix using alpha testing to indicate training data label accuracy.

Model Building: Decision Tree

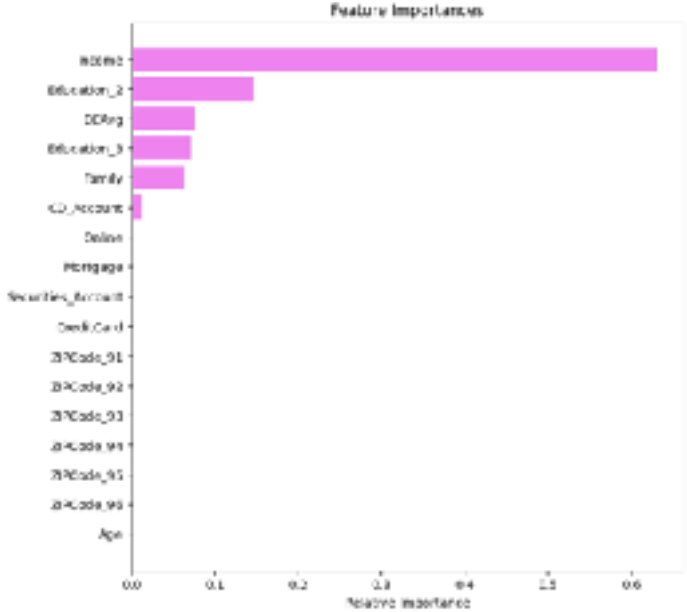
- Decision Tree on the Right showcases the categorization based on all variables present to cluster to the right variable

Income	0.308098
Family	0.259255
Education_2	0.166192
Education_3	0.147127
CCAvg	0.048798
Age	0.033150
CD_Account	0.017273
ZIPCode_94	0.007183
ZIPCode_93	0.004682
Mortgage	0.003236
Online	0.002224
Securities_Account	0.002224
ZIPCode_91	0.000556
ZIPCode_92	0.000000
ZIPCode_95	0.000000
ZIPCode_96	0.000000
CreditCard	0.000000



Final Model Pruning

- New Gini Tree indicates pruning is completed, With a new impurities list, there is a newer refined list of feature importances

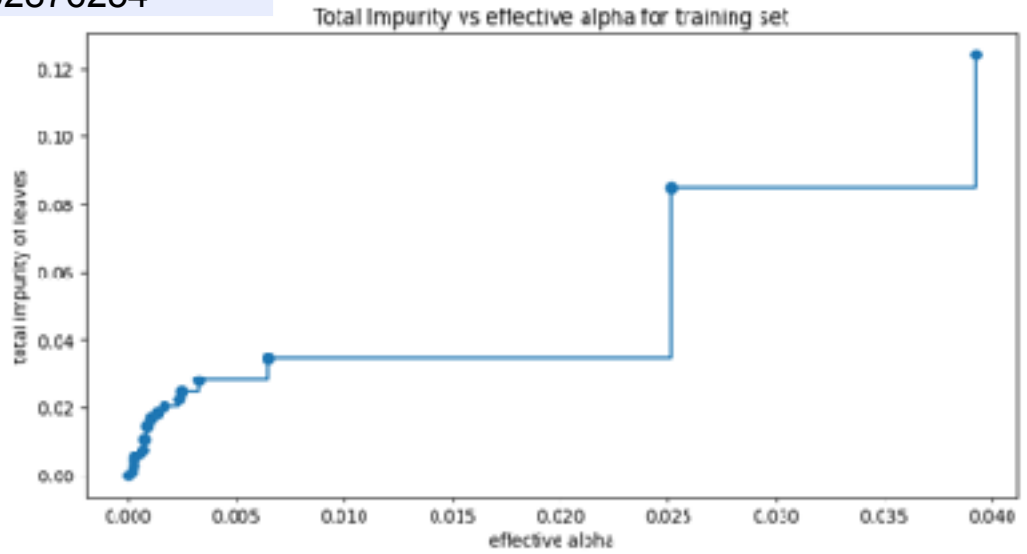




Model Performance Eval.

Table 1

index	Decision Tree sklearn	Decision Tree (Pre-Pruning)
Accuracy	0.986	0.986
Recall	0.9328859060402684	0.9328859060402684
Precision	0.9266666666666666	0.9266666666666666
F1	0.9297658862876254	0.9297658862876254



The graph indicates that the effective alpha value that would improve training performance is approximated to around 0.047

Actionable Insights

- Given the large correlation between income, personal loan, and CCAvg measures, AllLife Bank's asset customer demographic can only develop once loan marketing is targeted to individuals with lower average income and lower than average credit scores. Zip Code may influence ones financial standing as residence can be affected by CCAvg and may be used to finance loans, which can retain asset customers in the long run. These three variables provide great insight into marketable action for an increased asset customer demographic.

Great Learning

